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발표 자료

광운대학교 로봇학과
FAIR Lab

김한서

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이번 주 진행사항

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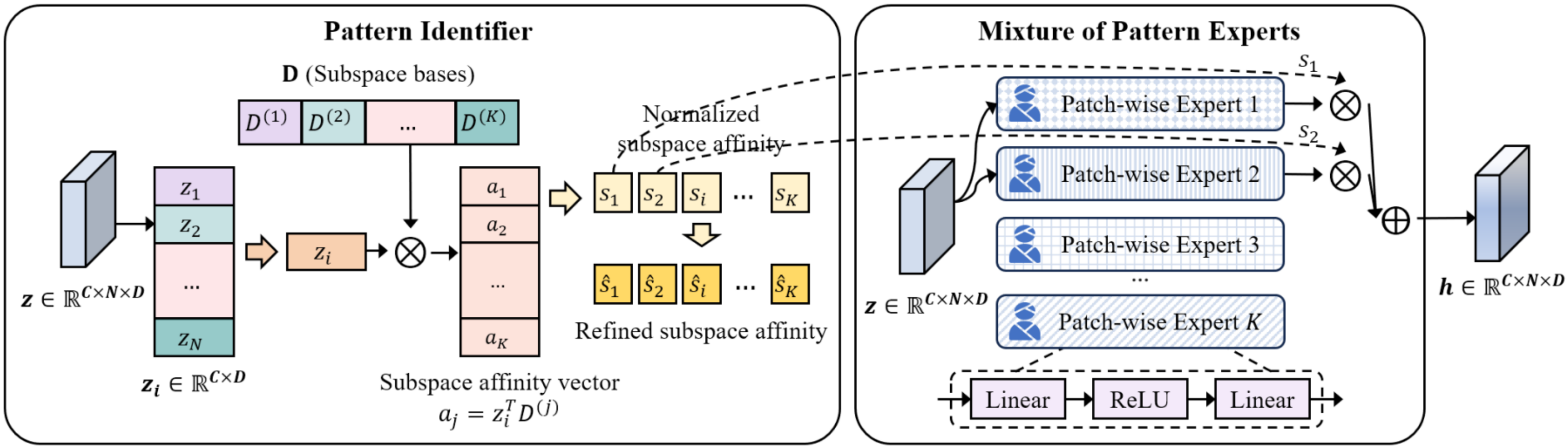
- Time-Frequency Pattern-Specific (TFPS)
  - 재현 실험 및 아이디어

모델 구조



- Pattern Identifier
  - 특징 벡터 z_i와 K개의 Subspace bases인 D를 행렬곱하여 유사도 계산, 이후 정규화 진행 $a_j = \mathbf{z_i^T} D^{(j)}$, $s_{ij} = \frac{|z_i^T D^{(j)}|_F^2 + \eta d}{\sum_j (|z_i^T D^{(j)}|_F^2 + \eta d)}$
  - 정규화된 유사도를 제곱하여 패턴 분류를 변별력있게 하고, 재정규화하여 $\widehat{s_{ij}} = \frac{s_{ij}^2 / \sum_i s_{ij}}{\sum_j (s_{ij}^2 / \sum_i s_{ij})}$ 도출
  - $\mathcal{L}_{sub} = KL(\hat{S} \parallel S) = \sum_i \sum_j \widehat{s_{ij}} \log \frac{\widehat{s_{ij}}}{s_{ij}}$, $\mathcal{L}_{PI} = R + \beta \mathcal{L}_{sub}$

- Mixture of Pattern Experts
  - 각 Expert에 특징 벡터 z를 입력해 Representation 생성, 정규화된 Top-k개의 유사도를 곱한 뒤
    이후 출력값을 가중합하여 최종 Representation h 생성 $G(s) = \mathrm{Softmax}(\mathrm{TopK}(s))$, $\mathbf{h} = \sum_{k=1}^{K} G(s) E_k(\mathbf{z})$

z_i: N개의 Patch 중 i번째 Patch
D: 학습 과정에서 찾아낸 시계열 패턴들의 기준점
a_j: i번째 Patch가 j번째 패턴과 얼마나 유사한지를 나타내는 점수

실험 세팅

- 사용한 모델: TFPS
- 재현 실험 데이터셋: ETTh1, ETTm1,

Experiment	ETTh1
Learning rate	5×10^{-4}
Epoch	100
Batch size	128
Loss function	MSE Loss
Seq_len	96
Pred_len	96/192/336/720
Layers	3
Expert	4~16

재현 실험 (ETTh1)

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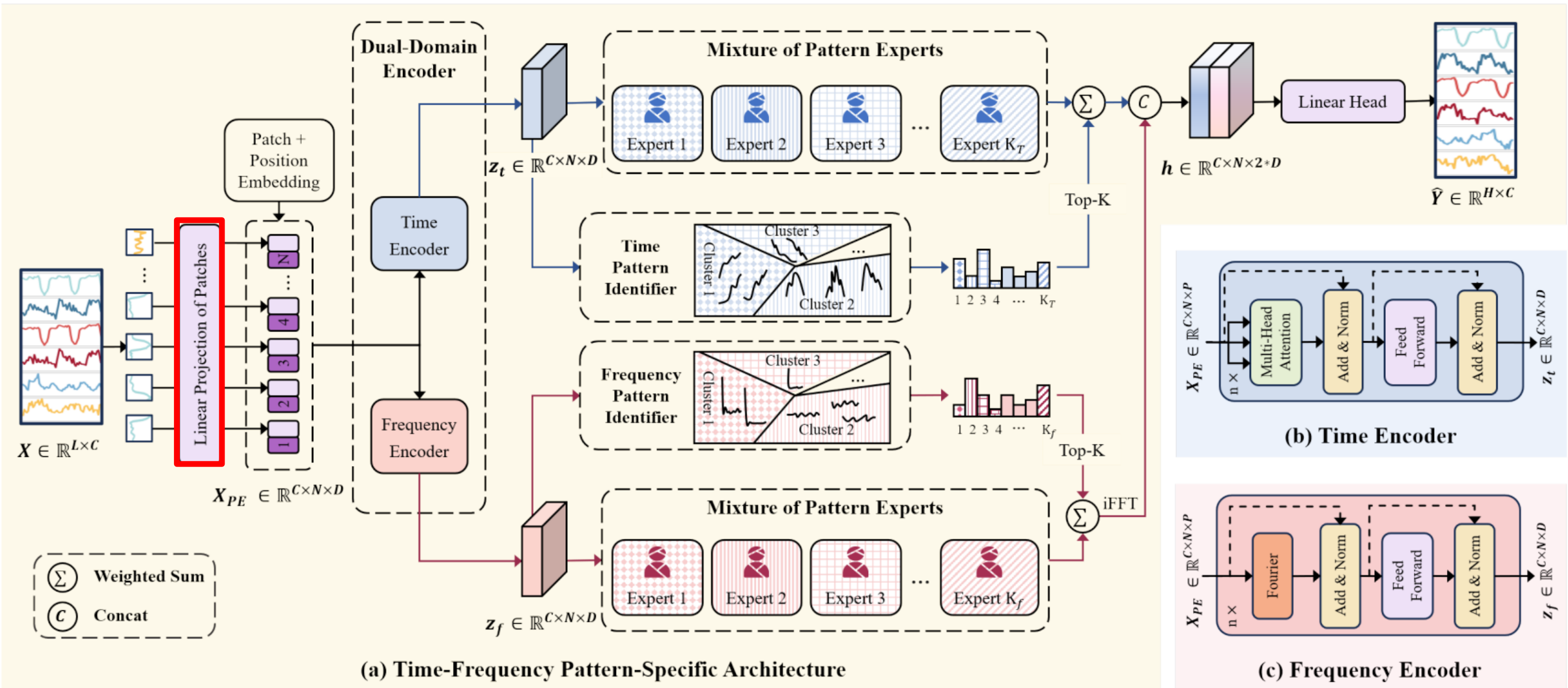
	ETTh1 Paper		ETTh1 Reproduction	
Prediction length	MSE	MAE	MSE	MAE
96	0.398	0.413	0.397	0.413
192	0.423	0.423	0.455	0.445
336	0.484	0.461	0.488	0.469
720	0.488	0.476	0.486	0.476

재현 실험 (ETTm1)

	ETTm1 Paper		ETTm1 Reproduction	
Prediction length	MSE	MAE	MSE	MAE
96	0.327	0.367	0.327	0.369
192	0.374	0.395	0.378	0.396
336	0.401	0.408	0.412	0.417
720	0.479	0.456	0.486	0.462

아이디어



- Linear Projection of Patches
  - 기존에는 고정된 Patch 길이를 사용하여, 모든 패턴을 동일한 Temporal Resolution으로만 관찰하기 때문에 단기적인 변동 또는 장기적인 추세가 제대로 파악이 안되어 정보 손실 발생
  - Patch 길이를 8, 16, 32와 같이 여러 Scale로 설정하여 병렬로 처리하고, Pattern Identifier가 패턴의 종류와 최적의 Scale을 동시에 판단하여 Expert에게 전달

이후 계획

- 아이디어 적용 후 실험

부록

주요 모델 성능 비교

Model		IMP.	TFPS (Our)		TSLANet (2024)		FITS (2024)		iTransformer (2024)		TFDNet-IK (2023)		PatchTST (2023)		TimesNet (2023)		DLinear (2023)		FEDformer (2022)	
Metric		MSE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	-1.1%	0.398	0.413	0.387	0.405	0.395	**0.403**	0.387	0.405	0.396	0.409	0.413	0.419	0.389	0.412	0.398	0.410	**0.385**	0.425
	192	4.8%	**0.423**	**0.423**	0.448	0.436	0.445	0.432	0.441	0.436	0.451	0.441	0.460	0.445	0.441	0.442	0.434	0.427	0.441	0.461
	336	1.8%	**0.484**	**0.461**	0.491	0.487	0.489	0.463	0.491	0.463	0.495	0.462	0.497	0.463	0.491	0.467	0.499	0.477	0.491	0.473
	720	3.0%	**0.488**	**0.476**	0.505	0.486	0.496	0.485	0.509	0.494	0.492	0.482	0.501	0.486	0.512	0.491	0.508	0.503	0.501	0.499
ETTh2	96	-2.0%	0.313	0.355	0.290	0.345	0.295	0.344	0.301	0.350	**0.289**	**0.337**	0.299	0.348	0.324	0.368	0.315	0.374	0.342	0.383
	192	-2.9%	0.405	0.410	**0.362**	**0.391**	0.382	0.396	0.380	0.399	0.379	0.395	0.383	0.398	0.393	0.410	0.432	0.447	0.434	0.440
	336	10.5%	**0.392**	**0.415**	0.401	0.419	0.416	0.425	0.424	0.432	0.416	0.422	0.424	0.431	0.429	0.437	0.486	0.481	0.512	0.497
	720	12.6%	**0.410**	**0.433**	0.419	0.439	0.418	0.437	0.430	0.447	0.424	0.441	0.429	0.445	0.433	0.448	0.732	0.614	0.467	0.476
ETTm1	96	4.1%	**0.327**	**0.367**	0.329	0.368	0.354	0.375	0.342	0.377	0.331	0.369	0.331	0.370	0.337	0.377	0.346	0.374	0.360	0.406
	192	2.6%	**0.374**	0.395	0.376	0.383	0.392	0.393	0.383	0.396	0.376	**0.381**	0.374	0.395	0.395	0.406	0.382	0.392	0.395	0.427
	336	4.2%	**0.401**	**0.408**	0.403	0.414	0.425	0.415	0.418	0.418	0.405	0.410	0.402	0.412	0.433	0.432	0.414	0.414	0.448	0.458
	720	-0.7%	0.479	0.456	**0.445**	0.438	0.486	0.449	0.487	0.457	0.471	**0.437**	0.466	0.446	0.484	0.458	0.478	0.455	0.491	0.479
ETTm2	96	6.9%	**0.170**	**0.255**	0.179	0.261	0.183	0.266	0.186	0.272	0.176	0.267	0.177	0.260	0.182	0.262	0.184	0.276	0.193	0.285
	192	7.1%	**0.235**	**0.296**	0.243	0.303	0.247	0.305	0.254	0.314	0.245	0.302	0.248	0.306	0.252	0.307	0.282	0.357	0.256	0.324
	336	4.6%	**0.297**	**0.335**	0.308	0.345	0.307	0.342	0.316	0.351	0.303	0.340	0.303	0.341	0.312	0.346	0.324	0.364	0.321	0.364
	720	3.6%	**0.401**	**0.397**	0.403	0.400	0.407	0.401	0.414	0.407	0.405	0.399	0.405	0.403	0.417	0.404	0.441	0.454	0.434	0.426
Exchange	96	12.7%	**0.083**	**0.205**	0.085	0.206	0.088	0.210	0.086	0.206	0.084	0.205	0.089	0.206	0.105	0.233	0.089	0.219	0.136	0.265
	192	11.2%	**0.174**	**0.297**	0.178	0.300	0.181	0.304	0.181	0.304	0.176	0.299	0.178	0.302	0.219	0.342	0.180	0.319	0.279	0.384
	336	10.4%	**0.310**	**0.398**	0.329	0.415	0.324	0.413	0.338	0.422	0.321	0.409	0.326	0.411	0.353	0.433	0.313	0.423	0.465	0.504
	720	-13.3%	1.011	0.756	0.850	0.693	0.846	0.696	0.853	0.696	**0.835**	**0.689**	0.840	0.690	0.912	0.724	0.837	0.690	1.169	0.826
Weather	96	15.6%	**0.154**	**0.202**	0.176	0.216	0.167	0.214	0.176	0.216	0.165	0.209	0.177	0.219	0.168	0.218	0.197	0.257	0.236	0.325
	192	10.6%	**0.205**	**0.249**	0.226	0.258	0.215	0.257	0.225	0.257	0.214	0.252	0.225	0.259	0.226	0.267	0.237	0.294	0.268	0.337
	336	9.1%	**0.262**	**0.289**	0.279	0.299	0.270	0.299	0.281	0.299	0.267	0.298	0.278	0.298	0.283	0.305	0.283	0.332	0.366	0.402
	720	4.1%	**0.344**	**0.342**	0.355	0.355	0.347	0.345	0.358	0.350	0.347	0.346	0.351	0.346	0.355	0.353	0.347	0.382	0.407	0.422
Electricity	96	14.6%	**0.149**	**0.236**	0.155	0.249	0.200	0.278	0.151	0.241	0.171	0.254	0.166	0.252	0.168	0.272	0.195	0.277	0.189	0.304
	192	12.0%	**0.162**	**0.253**	0.170	0.264	0.200	0.281	0.167	0.258	0.189	0.269	0.174	0.261	0.186	0.289	0.194	0.281	0.198	0.312
	336	0.2%	0.200	0.310	0.197	0.282	0.214	0.295	**0.179**	**0.271**	0.205	0.284	0.190	0.277	0.197	0.298	0.207	0.296	0.212	0.326
	720	7.2%	**0.220**	0.320	0.224	0.318	0.256	0.328	0.229	0.319	0.247	0.318	0.230	**0.312**	0.225	0.322	0.243	0.330	0.242	0.351
Traffic	96	21.1%	**0.427**	0.296	0.475	0.307	0.651	0.388	0.428	**0.295**	0.519	0.314	0.446	0.284	0.586	0.316	0.650	0.397	0.575	0.357
	192	17.7%	**0.445**	**0.298**	0.478	0.306	0.603	0.364	0.448	0.302	0.513	0.314	0.453	0.285	0.618	0.323	0.600	0.372	0.613	0.381
	336	17.0%	**0.459**	**0.307**	0.494	0.312	0.610	0.366	0.465	0.311	0.525	0.319	0.467	0.291	0.634	0.337	0.606	0.374	0.622	0.380
	720	15.1%	**0.496**	**0.313**	0.528	0.331	0.648	0.387	0.501	0.333	0.561	0.336	0.501	0.492	0.659	0.349	0.646	0.396	0.630	0.383
ILI	24	40.9%	**1.349**	**0.760**	1.749	0.898	3.489	1.373	2.443	1.078	1.824	0.824	1.614	0.835	1.699	0.871	2.239	1.041	3.217	1.246
	36	43.6%	**1.239**	**0.752**	1.754	0.912	3.530	1.370	2.455	1.086	1.699	0.813	1.475	0.859	1.733	0.913	2.238	1.049	2.688	1.074
	48	40.4%	**1.461**	**0.801**	2.050	0.984	3.671	1.391	3.437	1.331	1.762	0.831	1.642	0.880	2.272	0.999	2.252	1.064	2.540	1.057
	60	39.8%	**1.458**	**0.836**	2.240	1.039	4.030	1.462	2.734	1.155	1.758	0.863	1.608	0.885	1.998	0.974	2.236	1.057	2.782	1.136
1st Count			57		3		1		3		6		1		0		0		1	

Ablation Study

Table 2: Ablation study of TFPS components. The model variants in our ablation study include the following configurations across both time and frequency branches: (a) inclusion of the encoder, PI and MoPE; (b) PI replaced with Linear; (c) only the encoder. The best results are in **bold**.

Time Branch			Frequency Branch			ETTh1				ETTh2			
Encoder	PI	MoPE	Encoder	PI	MoPE	96	192	336	720	96	192	336	720
✓	✓	✓	✓	✓	✓	**0.398**	**0.423**	**0.484**	**0.488**	**0.313**	**0.405**	**0.392**	**0.410**
✓	✓	✓				0.401	0.459	0.486	0.492	0.318	0.409	0.400	0.428
✓	Linear	✓				0.401	0.451	0.494	0.509	0.325	0.411	0.400	0.434
✓						0.414	0.460	0.501	0.500	0.339	0.411	0.426	0.431
			✓	✓	✓	0.455	0.507	0.539	0.576	0.324	0.407	0.417	0.436
			✓	Linear	✓	0.503	0.535	0.558	0.583	0.398	0.446	0.457	0.444
			✓			0.552	0.583	0.591	0.594	0.371	0.426	0.418	0.463

- Time Branch와 Frequency Branch를 모두 사용한 기존 모델이 가장 좋은 성능을 보임
- Time Branch가 모델 성능에 큰 영향을 끼치는 것을 확인